

Sentiment Analysis Extension

<https://github.com/aiyshwary/Video-based-and-audio-based-sentiment-analysis>

OVERVIEW

A multimodal sentiment analysis project that independently extracts and analyses sentiment from both the video and audio recording, combining insights from two separate NLP and signal processing pipelines.

IMPACT

Built a dual-stream sentiment analysis pipeline that separates video and audio from a recording and independently processes each modality, enabling richer multimodal emotional understanding.

KEY DETAILS

1. Extracts audio track from video files and applies speech-to-text followed by NLP-based sentiment classification.
2. Processes video frames independently to infer visual sentiment cues from facial expressions and scene context.
3. Outputs positive, neutral, or negative sentiment labels with confidence scores per modality.
4. Designed for use cases including customer feedback analysis, social media monitoring, and presentation analysis.
5. Implemented using Hugging Face pretrained models and RapidMiner for pipeline orchestration.